Pathway Enrichment & Radiation Analysis

GO + KEGG + KEGG Module Enrichment Per Sub-Tissue, and Radiation-Focused Comparisons

Generated by Biomni | *2026-07-11*

What This Report Covers

Part 1: GO Biological Process, KEGG pathway, and KEGG module enrichment run separately for each significant sub-tissue's DEG list. Part 2: Radiation-focused analyses including volcano/MA plots, radiation type overlap (UpSet), dose-response analysis, and cross-comparison heatmaps. All results are in the Secondary_tissue_analysis folder alongside this PDF.

Part 1: Sub-Tissue Pathway Enrichment Results

Enrichment was run for each significant sub-tissue (from the Fisher's exact tests) in three comparisons: radiation_effect (Co-60 100 Gy), Cs137_100cGy, and Cs137_10cGy. Each sub-tissue's DEGs were tested in three directions (all, up, down) with three enrichment methods. Gene lists with fewer than 5 genes were skipped.

Summary statistics:

Count	What	Detail
21	Total enrichment runs	15 with GO results, 12 with KEGG, 4 with KEGG modules
32	CSV files generated	GO + KEGG + KEGG module result tables
31	Dot plot PNGs generated	One per significant enrichment result

Top GO Biological Process terms by sub-tissue (Co-60 100 Gy):

Sub-tissue	Genes	GO terms	Top enriched term	Theme
senescing_leaf	118	96	defense response to bacterium (padj=1.6e-13)	Immune/defense response dominates
cotyledon	67	32	cellular response to hypoxia (padj=1.7e-19)	Oxygen deprivation response
seedling_green_parts	26	39	indole glucosinolate metabolism (padj=1.3e-5)	Glucosinolate biosynthesis
sepal	36	10	response to singlet oxygen / jasmonic acid (padj=0.027)	Oxidative stress + JA signaling
root	26	7	cellular response to hypoxia (padj=2.3e-7)	Oxygen deprivation response
stem	7	14	regulation of immune response (padj=0.034)	Immune regulation
mature_pollen (down)	8	2	defense response (padj=0.04)	Limited enrichment

Key enrichment finding

> Different sub-tissues activate different biological processes in response to radiation. Senescing_leaf DEGs are dominated by immune/defense response genes (96 GO terms, top: defense response to bacterium, padj=1.6e-13). Cotyledon and root DEGs are enriched for hypoxia response (padj=1.7e-19 and 2.3e-7 respectively). Seedling_green_parts activates glucosinolate metabolism. This confirms that tissue-specific DEGs are not random -- they represent distinct biological pathways activated in different tissues.

KEGG pathway enrichment results:

Sub-tissue	Genes	KEGG	Top pathway	Theme
senescing_leaf	118	2	MAPK signaling pathway (padj=0.013)	Stress signaling
senescing_leaf	118	2	Sulfur metabolism (padj=0.050)	Sulfur-containing defense compounds
cotyledon	67	1	Plant-pathogen interaction (padj=2.2e-6)	Defense signaling
sepal	36	1	Plant-pathogen interaction (padj=0.027)	Defense signaling
mature_pollen (Cs137_100)	17	3	Ascorbate/aldarate metabolism (padj=0.019)	Antioxidant metabolism
mature_pollen (Cs137_100)	17	3	Inositol phosphate metabolism (padj=0.019)	Signaling
seedling_green_parts (Cs137_10)	9	1	Amino sugar/nucleotide sugar metabolism (padj=0.022)	Cell wall metabolism
seedling_green_parts (Cs137_100)	12	1	Diterpenoid biosynthesis (padj=0.034)	Secondary metabolism

KEGG module enrichment results:

Sub-tissue	Enriched module	Function
sepal (Co-60)	Monolignol biosynthesis (padj=0.038)	Lignin production -- cell wall strengthening
mature_pollen up (Co-60)	Pyrimidine biosynthesis UDP=>dTTP (padj=0.044)	DNA precursor synthesis
mature_pollen up (Co-60)	Citrate cycle first carbon oxidation (padj=0.044)	TCA cycle / energy metabolism
mature_pollen up (Co-60)	Glyoxylate cycle (padj=0.044)	Alternative carbon metabolism
mature_pollen down (Co-60)	Pectin degradation (padj=0.036)	Cell wall modification

KEGG interpretation

The KEGG results reveal tissue-specific pathway activation: senescing_leaf activates MAPK signaling and sulfur metabolism (defense compounds like glucosinolates), cotyledon and sepal activate plant-pathogen interaction pathways, and mature_pollen under Cs-137 shows antioxidant (ascorbate) and nucleotide sugar metabolism changes. The sepal-specific enrichment of monolignol biosynthesis (a KEGG module) suggests cell wall lignification is a tissue-specific radiation response.

Part 2: Radiation-Focused Analyses

2A. Radiation Type Comparison (UpSet + Overlap)

UpSet plots and pairwise overlap tables show which DEGs are shared vs unique across the 5 radiation conditions. Key findings:

Comparison	DEG counts	Shared	Jaccard	Interpretation
Co-60 100Gy vs Cs-137 100cGy	822 vs 229	84 shared	0.087	Low overlap -- different types activate mostly different genes
Cs-137 100cGy vs 10cGy	229 vs 87	63 shared	0.249	Highest overlap -- same type, dose-dependent
Co-60 100Gy vs Cs-137 10cGy	822 vs 87	34 shared	0.039	Very low overlap
Co-60 100Gy vs GCR 80cGy	822 vs 11	11 shared	0.013	GCR80 is a complete subset of Co-60 (11/11)
Cs-137 vs GCR	229 vs 2-11	0-1 shared	~0	GCR and Cs-137 share almost no DEGs

> **Universal vs radiation-type-specific DEGs**
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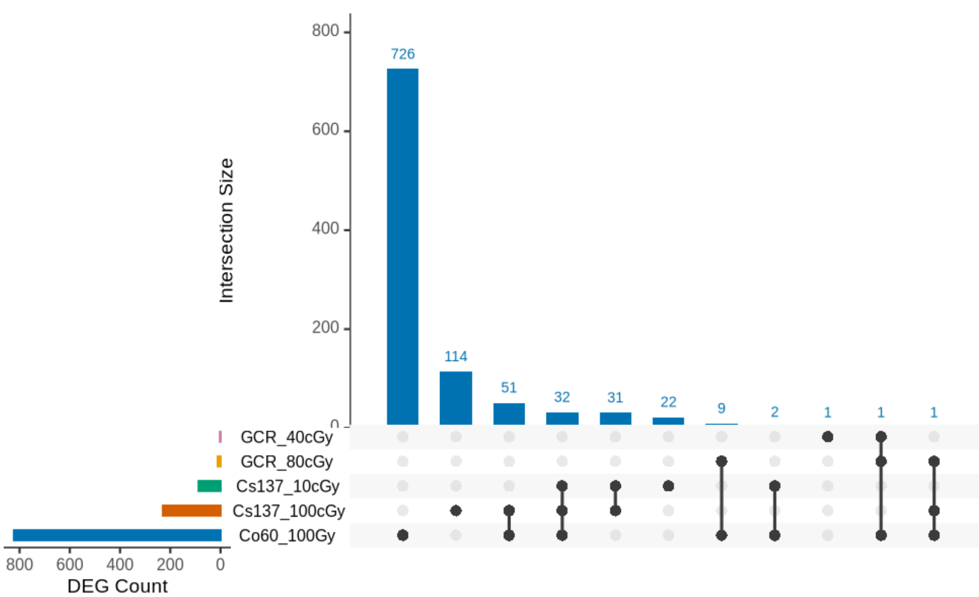
> 32 genes are 'universal' -- significant in all 3 gamma radiation types (Co-60 100 Gy, Cs-137 100 cGy, Cs-137 10 cGy). These represent the core radiation response. 736 genes are Co-60 specific, 114 are Cs-137 100 cGy specific, and 22 are Cs-137 10 cGy specific. GCR 80 cGy's 11 DEGs are entirely contained within the Co-60 DEG set, suggesting GCR activates a subset of the gamma radiation response.



UpSet plot: DEG overlap across all 5 radiation types.

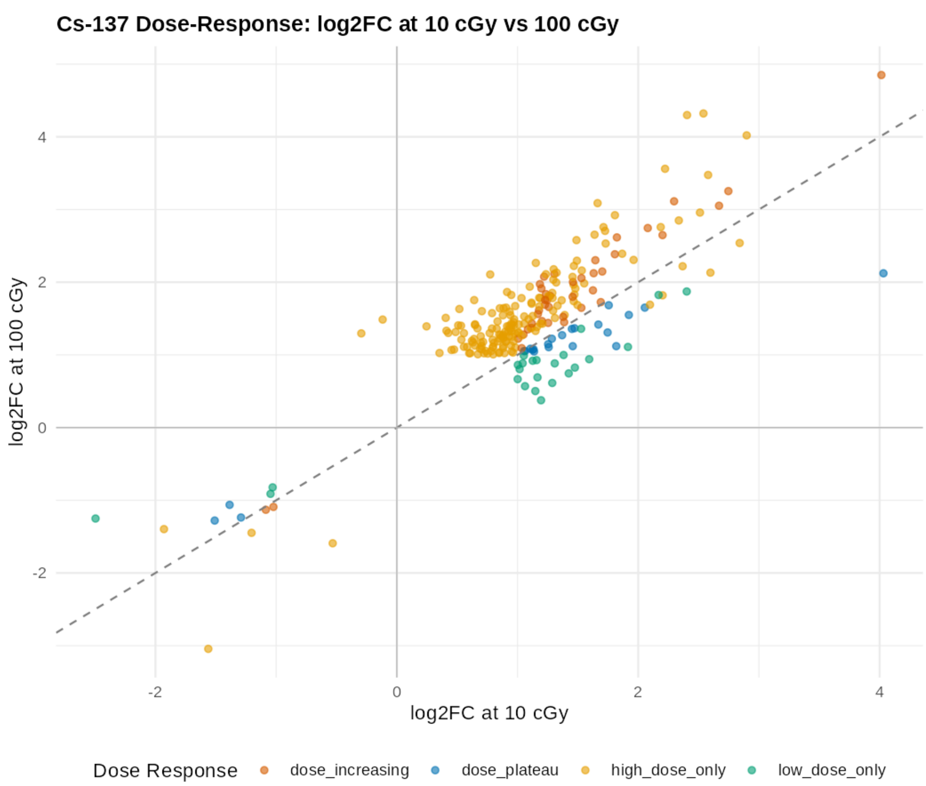
2B. Dose-Response Analysis

For Cs-137 gamma radiation, we compared DEG status at 10 cGy vs 100 cGy to identify dose-responsive genes. For GCR, we compared 40 cGy vs 80 cGy.

Category	Genes	Definition
Dose-increasing	40	Significant at both doses, |log2FC| increases with dose
Dose-plateau	23	Significant at both doses, |log2FC| similar at both doses
High-dose-only	166	Significant only at 100 cGy (not at 10 cGy)
Low-dose-only	24	Significant only at 10 cGy (not at 100 cGy)

Dose-response statistics

63 genes are dose-responsive (significant at both Cs-137 doses). Of these, 40 show dose-increasing response (larger |log2FC| at 100 cGy) and 23 show a plateau (similar magnitude at both doses). The paired Wilcoxon test confirms that |log2FC| is significantly higher at 100 cGy than 10 cGy ($p = 0.0004$). Median |log2FC| increases from 1.37 (10 cGy) to 1.56 (100 cGy). For GCR, only 1 gene is shared between 40 cGy and 80 cGy, with 10 high-dose-only and 1 low-dose-only -- too few for statistical testing.

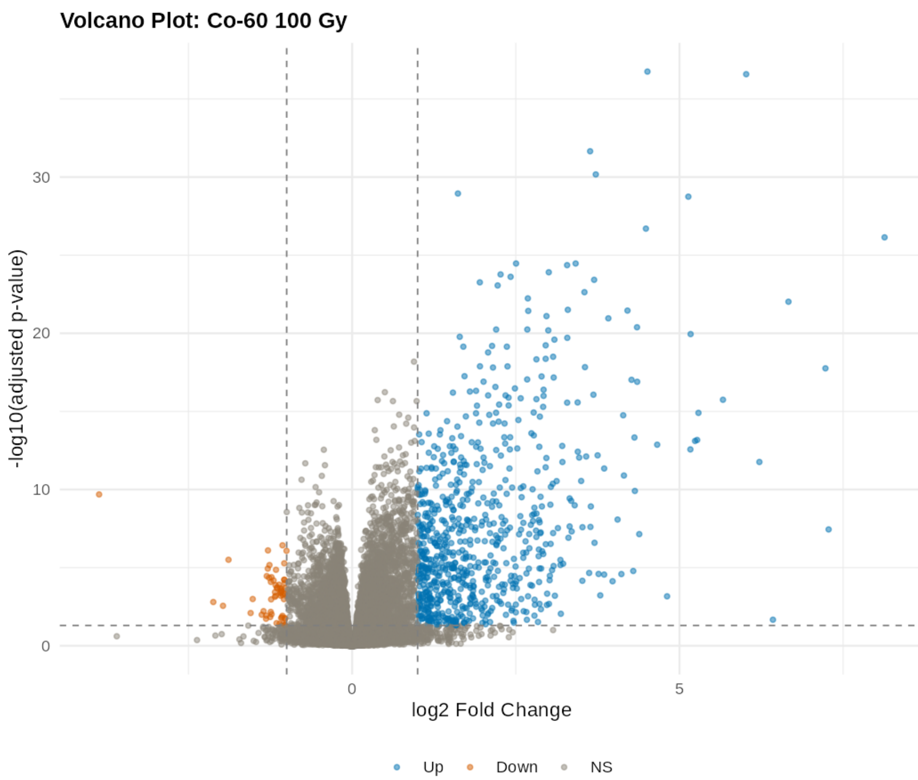


Cs-137 dose-response scatter: log2FC at 10 cGy vs 100 cGy. Points above the diagonal have larger response at high dose.

2C. Volcano and MA Plots

Volcano plots (log2FC vs -log10 padj) and MA plots (log2FC vs baseMean) were generated for all 6 comparisons. Key observations from the volcano plots:

Comparison	DEGs	Direction split	Observation
Co-60 100 Gy	822	764 up vs 58 down	Largest DEG count, mostly upregulated
sog1-1 Interaction	433	76 up vs 357 down	Direction reversal confirms SOG1 role
Cs-137 100 cGy	229	220 up vs 9 down	Almost entirely upregulated
Cs-137 10 cGy	87	79 up vs 8 down	Same pattern, fewer DEGs
GCR 80 cGy	11	11 up vs 0 down	All upregulated, very few DEGs
GCR 40 cGy	2	1 up vs 1 down	Too few for pattern assessment



Volcano plot: Co-60 100 Gy. Blue = upregulated, orange = downregulated, gray = not significant.

2D. Cross-Comparison Heatmap

A heatmap of 321 genes (297 significant in >= 2 comparisons + 50 top by |log2FC| from Co-60) shows how the same genes respond across all 6 radiation conditions. Genes are row-clustered to reveal groups with similar response patterns.

> **Heatmap interpretation**
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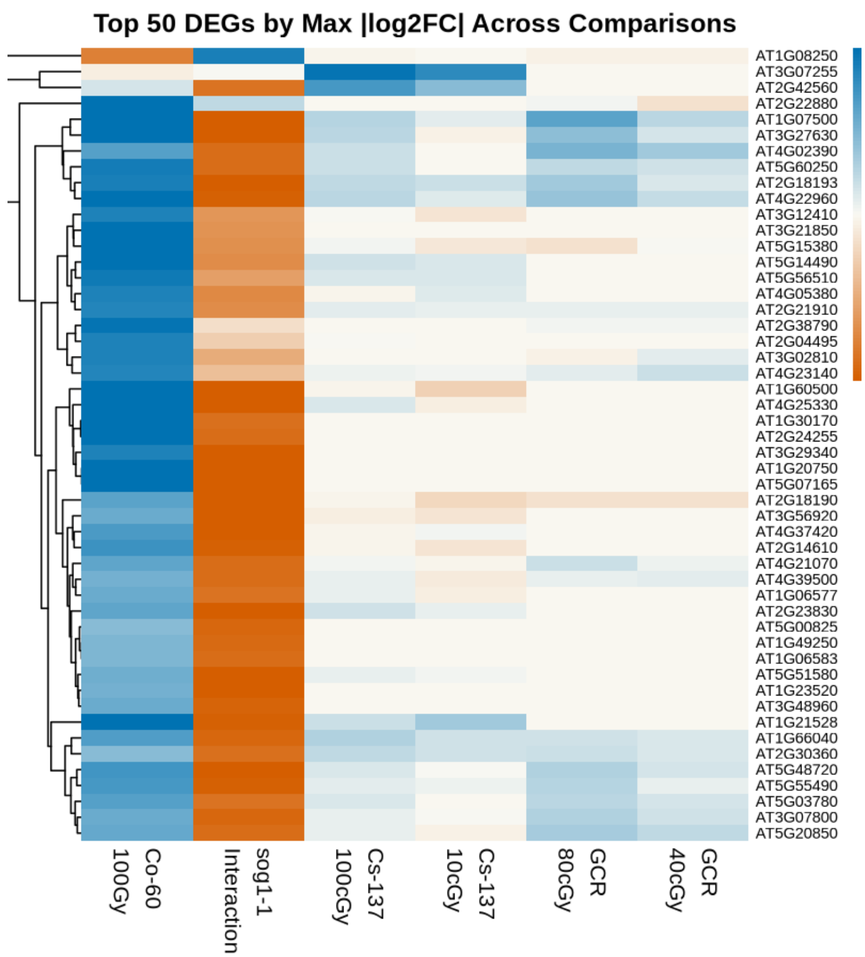
> The heatmap reveals distinct gene clusters: (1) a large cluster upregulated across all gamma types (the universal radiation response), (2) a cluster upregulated by Co-60 but downregulated by the sog1-1 interaction (SOG1-dependent genes), (3) a cluster specific to Cs-137 (mostly root/seed genes), and (4) GCR-responsive genes that overlap with Co-60. The genotype_interaction column shows a clear mirror image of the radiation_effect column for many genes, visually confirming the direction reversal.



Top 50 DEGs by max |log2FC| across all 6 comparisons. Blue = upregulated, orange = downregulated.

Part 3: File Inventory

All results are in the Secondary_tissue_analysis folder. Here is what each file contains:

File(s)	Count	Content
enrichment_results/	32 CSVs + 31 PNGs	GO + KEGG + KEGG module results per sub-tissue
enrichment_summary_all.csv	1 CSV	Summary of all 21 enrichment runs
subtissue_enrichment_analysis.R	1 R script	Reusable script to re-run all enrichment
code_walkthrough_pathway_enrichment.pdf	12-page PDF	Line-by-line explanation of your original enrichment code
fig_rad_volcano_*.png	6 PNGs	Volcano plots for each comparison
fig_rad_ma_*.png	6 PNGs	MA plots for each comparison
fig_rad_upset_*.png	2 PNGs	UpSet plots (all 5 types + gamma only)
radiation_type_overlap_table.csv	1 CSV	Pairwise DEG overlap + Jaccard indices
deg_universal_gamma.csv	1 CSV	32 genes significant in all 3 gamma types
deg_Co60_specific.csv	1 CSV	736 genes unique to Co-60
deg_Cs137_*_specific.csv	2 CSVs	Genes unique to each Cs-137 dose
dose_response_Cs137.csv	1 CSV	All genes with dose-response category
dose_responsive_genes_Cs137.csv	1 CSV	63 genes significant at both Cs-137 doses
dose_response_GCR.csv	1 CSV	GCR dose-response categories
fig_rad_dose_response_*.png	2 PNGs	Dose-response scatter plots
fig_rad_cross_comparison_heatmap.png	1 PNG	321-gene heatmap across all comparisons
fig_rad_top50_heatmap.png	1 PNG	Top 50 gene heatmap
cross_comparison_log2fc_matrix.csv	1 CSV	log2FC matrix for all heatmap genes

Total deliverables in this folder

402 files total (35 MB), including: 6 PDFs, 63 PNG figures, 28+ CSV data tables, 2 R scripts, and 246 hierarchical tissue CSVs in the tissue_specific_degs_v2/ subfolder. Everything is downloadable from the Secondary_tissue_analysis folder.